

Semantic Reading Order Reconstruction for Local-First Research Notebooks

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Abstract

Knowledge work depends on the careful integration of heterogeneous sources, yet most tools treat documents as opaque images. Our system reconstructs reading order from positioned glyph runs, detects columns through gap analysis, and preserves typographic structure. Empirically, the approach recovers paragraphs with high fidelity on a corpus of two-column conference papers and textbooks.

1 Introduction

Researchers routinely read hundreds of papers, and the act of excerpting a passage should preserve its information rather than flatten it into a screenshot. Knowledge work depends on the careful integration of heterogeneous sources, yet most tools treat documents as opaque images. We argue that a local-first architecture offers durable ownership of research notes while enabling real-time collaboration between peers.

Conflict-free replicated data types provide strong eventual consistency without a central coordinator, which makes them a natural foundation. Our system reconstructs reading order from positioned glyph runs, detects columns through gap analysis, and preserves typographic structure.

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2 Related Work

Prior systems for semantic extraction rely on heuristics over geometric features of text runs. Empirically, the approach

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3 Method

We model a page as a set of positioned items and cluster them into columns by detecting wide vertical gutters. Our system reconstructs reading order from positioned glyph runs, detects columns through gap analysis, and preserves typographic structure.

$$\sum \alpha_i x_i^2 \leq \beta \quad (1)$$

The objective above is minimized with respect to the parameters, subject to the normalization constraint. Conflict-free replicated data types provide strong eventual consistency without a central coordinator, which makes them a natural foundation.

4 Evaluation

We argue that a local-first architecture offers durable ownership of research notes while enabling real-time collaboration between peers. Our system reconstructs reading order from positioned glyph runs, detects columns through gap analysis, and preserves typographic structure. Knowledge work depends on the careful integration of heterogeneous sources, yet most tools treat documents as opaque images. Conflict-free replicated data types provide strong eventual consistency without a central coordinator, which makes them a natural foundation.

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5 Conclusion

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References

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